

SOC Home Lab Case Study

Project Title:

Splunk Authentication Event
Detection and Alerting

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Date: July 29, 2026 – August 1, 2026

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1. Executive Summary

This project demonstrates the creation of a basic Security Operations Center (SOC) home lab using Splunk Enterprise. The objective was to generate authentication events from a Linux system (Metasploitable), forward the logs to Splunk, create a detection search for failed login attempts, configure an alert, and validate that the alert was triggered successfully.

The project simulates a real-world SOC monitoring workflow, from log collection through detection and alerting, providing hands-on experience with SIEM operations and incident detection.

2. Project Objective

The project involved:

- * Installing and configuring Splunk Enterprise on the host machine.
 - * Installing and configuring the Splunk Universal Forwarder to collect Linux authentication logs.
 - * Using Metasploitable 2 as the vulnerable target machine to generate authentication events.
 - * Simulating failed SSH login attempts to produce security-relevant logs.
 - * Forwarding authentication logs from Metasploitable to Splunk for analysis.
 - * Creating a detection search in Splunk to identify failed authentication attempts.
 - * Configuring an alert to notify when suspicious authentication activity was detected.
 - * Validating that the alert triggered successfully after the simulated attack.
 - * Documenting the complete deployment, troubleshooting process, and results.
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3. Lab Environment

Component	Description
Host Machine	Windows Laptop
Virtualization Platform	Oracle VirtualBox
SIEM Platform	Splunk Enterprise
Log Forwarder	Splunk Universal Forwarder
Attack Machine	Kali Linux
Target Machine	Kali Linux
Operating Systems	Windows, Kali Linux, Metasploitable 2
Log Source	Linux Authentication Logs (/var/log/auth.log)
Forwarding Port	TCP Port 9997
Detection Method	Splunk Search Processing Language
Project Focus	Authentication Monitoring, Log Analysis, and Alert Creation

4. Tools Used

Tool	Purpose
Splunk Enterprise	Security Information and Event Management (SIEM) platform used to collect, search, analyze, and monitor security logs.
Splunk Universal Forwarder	Forwarded Linux authentication logs from the target machine to Splunk Enterprise.
Kali Linux	Used to simulate failed SSH login attempts and generate authentication events.
Metasploitable 2	Vulnerable Linux virtual machine used as the target system for generating authentication logs.
Oracle VirtualBox	Virtualization platform used to host Kali Linux and Metasploitable 2 virtual machines.
SSH	Used to attempt remote logins from Kali Linux to Metasploitable 2, generating authentication events.
Linux Authentication Logs (/var/log/auth.log)	Primary log source monitored by Splunk for failed authentication attempts.
Splunk Search Processing Language (SPL)	Used to search, analyze, and detect failed authentication events.

5. Network Architecture

The SOC home lab consisted of a Windows host machine running Oracle VirtualBox, which hosted both the Kali Linux attack machine

and the Metasploitable 2 target machine. Splunk Enterprise was installed on the Windows host to act as the Security Information and Event Management (SIEM) platform. The Splunk Universal Forwarder was configured on the Metasploitable 2 virtual machine to forward Linux authentication logs to Splunk Enterprise over TCP port 9997.

Kali Linux was used to perform failed SSH login attempts against Metasploitable 2. These authentication events were recorded in the Linux authentication logs, forwarded to Splunk Enterprise, and analyzed using Splunk Search Processing Language (SPL). A detection rule was configured to identify failed login attempts and trigger an alert.

6. Implementation

Step 1: Metasploitable 2 Setup

- Imported and configured the Metasploitable 2 virtual machine in VirtualBox.
- Verified that the system was running correctly and accessible on the virtual network.
- Confirmed that SSH and other vulnerable services were available for testing.

Step 2: Splunk Enterprise Installation

- Installed Splunk Enterprise on an Ubuntu virtual machine.
- Completed the initial web-based configuration and created an administrator account.

Step 3: Splunk Universal Forwarder Configuration

- Installed and configured the Splunk Universal Forwarder.
- Configured it to monitor Linux authentication logs and forward them to the Splunk Enterprise server.

Step 4: Log Collection Verification

- Confirmed that authentication logs from the monitored system were successfully received and indexed in Splunk.
- Verified that the logs were searchable using the Search & Reporting application.

Step 5: Kali Linux Attack Simulation

- Used Kali Linux to perform SSH login attempts against the Metasploitable 2 virtual machine.
- Generated both successful and failed authentication events to simulate attacker activity.

Step 6: Detection Development

- Created an SPL query to identify failed SSH authentication attempts.
- Tested and refined the query to ensure accurate detection.

Step 7: Alert Configuration

- Configured a Splunk alert based on the detection query.
- Set trigger conditions to notify when multiple failed login attempts occurred.

Step 8: Validation

- Confirmed that the simulated attack generated authentication logs.
- Verified that the SPL query detected the events correctly.
- Validated that the Splunk alert triggered as expected.

7. Detection Logics

The detection logic was designed to identify failed SSH authentication attempts generated during the attack simulation. Authentication logs forwarded from the monitored Linux system were continuously indexed by Splunk Enterprise.

A Splunk Search Processing Language (SPL) query was used to search for failed password events within the authentication logs. When failed login attempts matching the detection criteria were identified, Splunk generated an alert for further investigation.

spl

```
index=* sourcetype=linux_secure "Failed password"
```

Detection Criteria

- * Log Source: Linux authentication logs
- * Event Type: Failed SSH login attempts
- * Detection Method: Splunk Search Processing Language (SPL)
- * Alert Trigger: When failed authentication events were detected

8. Alert Configuration

A Splunk alert was created based on the detection query for failed SSH authentication attempts. The alert was configured to monitor authentication events in real time and trigger whenever the search identified failed login attempts that matched the defined criteria.

Alert Details

Setting	Configuration
Alert Name	Linux Failed Authentication Detection
Trigger Type	Real-time
Detection Method	SPL Query
Severity	Medium
Event Source	Linux Authentication Logs

9. Alert Validation

To validate the detection, failed SSH login attempts were generated from the Kali Linux attack machine against the Metasploitable 2 target machine. The authentication events were successfully forwarded to Splunk Enterprise through the Splunk Universal Forwarder. The events were indexed, detected by the SPL query, and the configured alert was triggered successfully, confirming that the monitoring and detection workflow was functioning as intended.

Search | Splunk 10.4.1

http://localhost:8000/en-US/app/search/search?q=search index%3D* sourcetype%3Dsyslog ("authentication failure" OR "incorrect password")

Time range: Last 60 minutes

4 events (7/31/26 2:44:00.000 PM to 7/31/26 3:44:07.000 PM) No Event Sampling

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # COMMAND 1
- # date_hour 1
- # date_mday 1
- # date_minute 2
- # date_month 1
- # date_second 4
- # date_wday 1
- # date_year 1
- # date_zone 1
- # euid 1
- # index 1

Time	Event
7/31/26 2:56:45.000 PM	Jul 31 14:56:45 vtcsec sudo: marlinspike : 3 incorrect password attempts ; TTY=pts/0 ; PWD=/home/marlinspike ; USER=root ; COMMAND=/bin/ls host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:56:30.000 PM	Jul 31 14:56:10 vtcsec sudo: pam_unix(sudo:auth): authentication failure; logname= uid=1000 euid=0 tty=/dev/pts/0 ruser=marlinspike rhost= user=marlinspike host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:55:48.000 PM	Jul 31 14:55:48 vtcsec sudo: marlinspike : 3 incorrect password attempts ; TTY=pts/0 ; PWD=/home/marlinspike ; USER=root ; COMMAND=/bin/ls host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:55:36.000 PM	Jul 31 14:55:36 vtcsec sudo: pam_unix(sudo:auth): authentication failure; logname= uid=1000 euid=0 tty=/dev/pts/0 ruser=marlinspike rhost= user=marlinspike host = vtcsec source = /var/log/auth.log sourcetype = syslog

Search | Splunk 10.4.1

http://localhost:8000/en-US/app/search/search?q=search index%3D* sourcetype%3Dsyslog&display=page

Time range: Last 60 minutes

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

Hide Fields All Fields

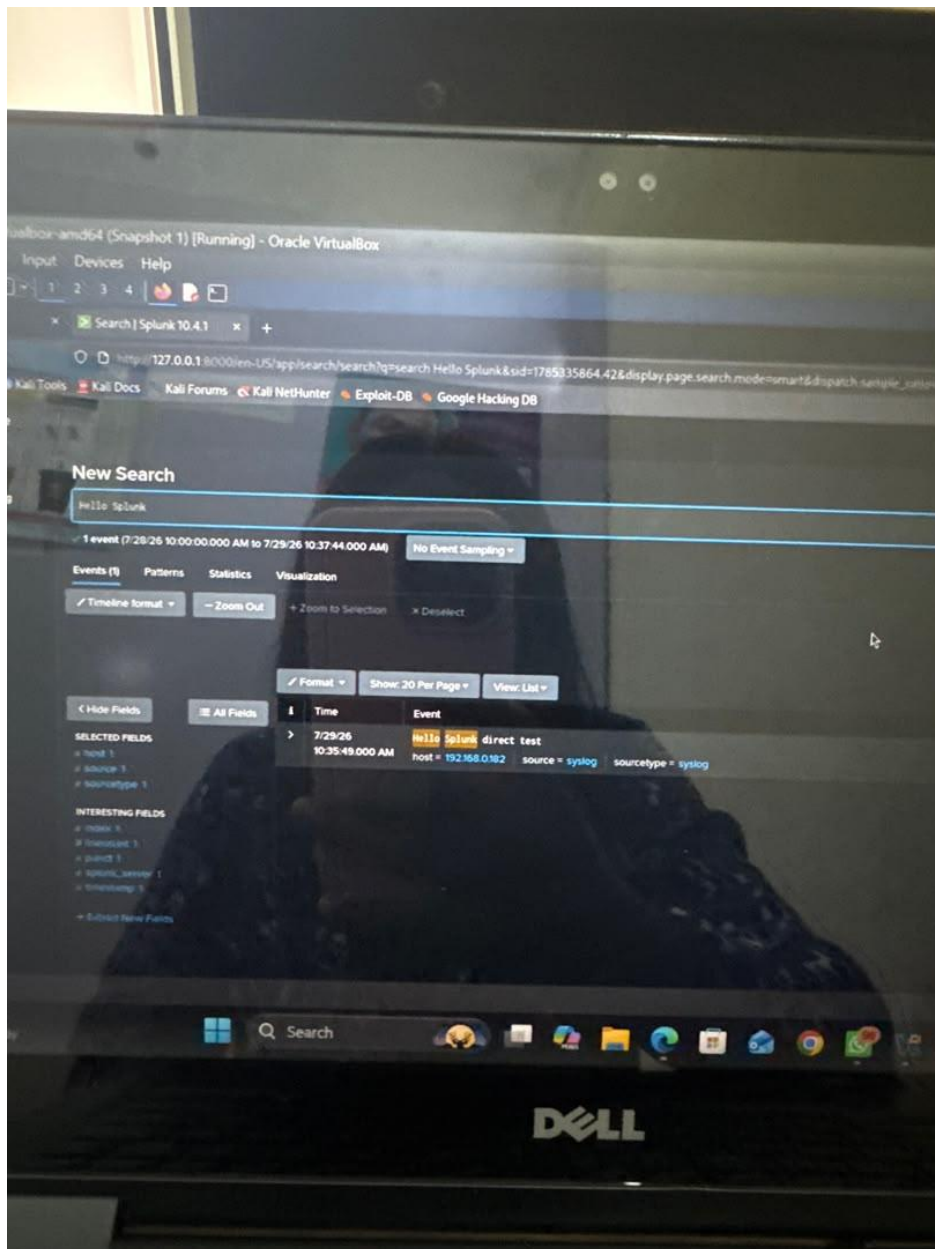
SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # COMMAND 2
- # date_hour 1
- # date_mday 1
- # date_minute 8
- # date_month 15
- # date_wday 1
- # date_year 1
- # date_zone 1
- # index 1
- # linecount 1
- # process 1
- # punct 8
- # PWD 1
- # splunk_server 1
- # timeendpos 1
- # timeoffset 1

Time	Event
7/31/26 2:57:16.000 PM	Jul 31 14:57:16 vtcsec sudo: pam_unix(sudo:session): session closed for user root host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:57:16.000 PM	Jul 31 14:57:16 vtcsec sudo: pam_unix(sudo:session): session opened for user root by (uid=0) host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:57:16.000 PM	Jul 31 14:57:16 vtcsec sudo: marlinspike : TTY=pts/0 ; PWD=/home/marlinspike ; USER=root ; COMMAND=/bin/ls host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:56:45.000 PM	Jul 31 14:56:45 vtcsec sudo: marlinspike : 3 incorrect password attempts ; TTY=pts/0 ; PWD=/home/marlinspike ; USER=root ; COMMAND=/bin/ls host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:56:30.000 PM	Jul 31 14:56:10 vtcsec sudo: pam_unix(sudo:auth): authentication failure; logname= uid=1000 euid=0 tty=/dev/pts/0 ruser=marlinspike rhost= user=marlinspike host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:55:48.000 PM	Jul 31 14:55:48 vtcsec sudo: marlinspike : 3 incorrect password attempts ; TTY=pts/0 ; PWD=/home/marlinspike ; USER=root ; COMMAND=/bin/ls host = vtcsec source = /var/log/auth.log sourcetype = syslog
7/31/26 2:55:36.000 PM	Jul 31 14:55:36 vtcsec sudo: pam_unix(sudo:auth): authentication failure; logname= uid=1000 euid=0 tty=/dev/pts/0 ruser=marlinspike rhost= user=marlinspike host = vtcsec source = /var/log/auth.log sourcetype = syslog



10. Challenges Encountered

During the implementation of the SOC home lab, several challenges were encountered:

- Configuring the Splunk Universal Forwarder to successfully send authentication logs to Splunk Enterprise.
- Troubleshooting connectivity issues between the virtual machines.
- Ensuring that Linux authentication logs were correctly indexed in Splunk.
- Verifying that the detection query returned the expected events.
- Configuring the alert to trigger correctly after failed SSH login attempts.

These challenges were resolved through troubleshooting, configuration adjustments, and repeated testing until the environment functioned as expected.

11. Lessons Learned

This project provided practical experience in:

- Deploying and configuring Splunk Enterprise.
 - Configuring the Splunk Universal Forwarder for log collection.
 - Monitoring Linux authentication logs.
 - Writing and testing SPL detection queries.
 - Creating and validating security alerts.
 - Troubleshooting SIEM deployment and log forwarding issues.
 - Understanding the workflow of a Security Operations Center (SOC).
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12. Skills Demonstrated

This project demonstrates the following technical skills:

- Security Information and Event Management (SIEM)
- Splunk Enterprise Administration
- Splunk Universal Forwarder Configuration
- Linux Log Collection and Monitoring
- Security Log Analysis
- Splunk Search Processing Language (SPL)
- Authentication Event Detection
- Alert Creation and Validation
- VirtualBox Virtualization

- Linux System Administration
 - Incident Detection and Monitoring
 - Troubleshooting and Problem Solving
 - Technical Documentation
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13. Conclusion

This project successfully demonstrated the implementation of a SOC home lab using Splunk Enterprise. Authentication logs were collected from a Linux system, indexed into Splunk, and analyzed using SPL queries. Failed SSH login attempts generated from Kali Linux against the Metasploitable 2 target machine were successfully detected, and the configured alert was triggered as expected.

The project provided practical experience in SIEM deployment, log forwarding, security monitoring, detection engineering, alert configuration, and troubleshooting. It also strengthened my understanding of SOC operations and the role of Splunk in monitoring and investigating security events.